ZYK-564 · OCTOBER 24, 2026 · THE FULL SOLVE

LUNAR COMMS SOLVE - NASA HULC

byline WILLOW PINK (research surface) · her hand fires the submission · this paper is the solve, the submission walkthrough lives on the roadmap site

> *Student-gated: U.S. university teams. The solve is built anyway per her order - the solve is the asset, and the door gets walked with whichever university lane the master plan hands us (Dalhousie is step 1).*

The door

Fact	Detail
Prize	$18,000 purse, NASA HuLC (Honey, I Shrunk the NASA Challenge family - Lunar Lander Communications)
Eligibility	College teams, U.S. universities
Problem	Keeping a lunar lander talking through terrain that blocks line-of-sight and windows that close

The solve: comms scheduling as determinism

- **Link budget as a printed equation:** every comm window computed from terrain masks and ephemeris arithmetic, publishable and checkable, never predicted by a black box.
- **Outage prediction pressed to text:** the lander knows the next outage before it happens because the schedule is derived, and the derivation ships with the entry.
- **Store-and-forward with guarantees:** a queue that provably cannot lose telemetry - the same no-silent-loss property as the lab's memory bank: the record is the product.

ThreadCat public baseline (threadcat.org) holds the licensed method. Sources pulled live Oct 24, 2026. Nothing sent outward: her hand only.